ViT-Enhanced Critic for Terrain-Adaptive Bipedal Locomotion via Teacher-Student Reinforcement Learning

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***Abstract*—Bipedal robots hold substantial promise for navigating human-centric environments, yet their inherent underactuation renders terrain-adaptive locomotion control a persistent challenge. While recent teacher-student reinforcement learning (RL) frameworks have advanced multi-terrain locomotion, the terrain encoding modules in existing approaches—typically based on Multilayer Perceptrons (MLPs) or Mixture-of-Experts (MoE) models—struggle to capture the spatial correlations embedded in terrain heightmaps. In this paper, we propose a Vision Transformer (ViT)-enhanced critic architecture that replaces conventional terrain encoders with a lightweight ViT module. The ViT encoder tokenizes a 12×12 local terrain heightmap into patch embeddings, applies self-attention to model spatial dependencies among terrain patches, and aggregates the global terrain representation via a learnable `[CLS]` token. This terrain feature is fused with proprioceptive and privileged observations to produce value estimates that better reflect the traversability of the surrounding environment. We integrate this architecture into a Proximal Policy Optimization (PPO) pipeline with conservative clipping and validate it through simulation experiments across five terrain types and real-world deployment on a bipedal robot. In simulation across five terrain types, the proposed method reduces velocity tracking error by 5.0% and height tracking error by 8.5% compared to the state-of-the-art VMTS baseline, while real-world deployment on a Limx Dynamics P1 robot validates stair climbing and multi-terrain traversal.**

***Index Terms*—Bipedal robots, reinforcement learning, Vision Transformer, terrain-adaptive locomotion, teacher-student learning**

I. Introduction

Bipedal robots can potentially operate in human-centric environments ranging from logistics warehouses to disaster sites [1]–[3]. Traversing stairs, slopes, and uneven ground, however, demands real-time terrain perception coupled with adaptive gait generation—a task complicated by the inherent underactuation of point-foot bipedal platforms [4]. Robust traversability assessment [26] and efficient trajectory planning on uneven terrain [29], [30] have been studied extensively for wheeled platforms, yet adapting these ideas to bipedal robots with point-foot contacts remains open.

The teacher-student reinforcement learning (RL) paradigm has become a practical solution for deploying locomotion policies trained in simulation [5]–[7]. A privileged teacher, with access to terrain heightmaps and physical parameters, first learns a robust policy inside a simulator [8]–[10]. A deployable student then distills this policy using only onboard sensors.

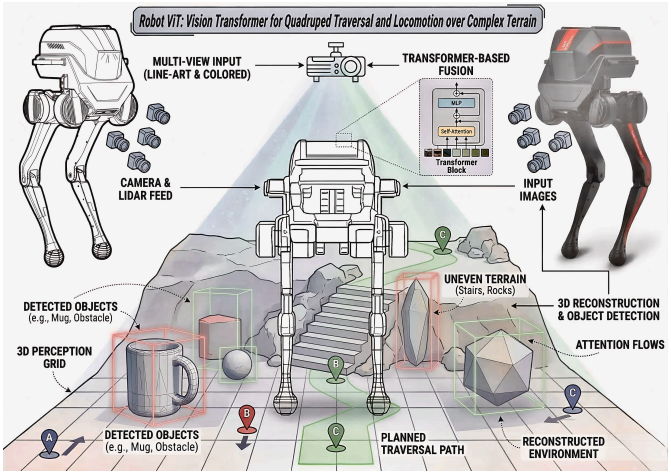


Fig. 1: Conceptual illustration of Vision Transformer-based terrain perception for legged robot locomotion over complex terrain. A Transformer-based fusion module processes multi-view terrain information to enable adaptive traversal across stairs, rocks, and uneven surfaces.

The bottleneck in this pipeline lies in *how the terrain heightmap is encoded*. Standard MLPs flatten the 12×12 heightmap into a 144-dimensional vector, erasing spatial adjacency [7], [8]. VMTS [14] addresses this with a Mixture-of-Experts (MoE) model that routes different terrain types to specialized sub-networks; yet MoE still requires discrete terrain classification and cannot model fine-grained spatial correlations *within* a single heightmap patch.

We observe that adjacent cells in a terrain heightmap are *not* independent: a stair edge next to flat ground changes traversability in a way that depends on the spatial context. Self-attention can capture exactly this kind of pairwise dependency without hand-designed terrain categories. Based on this observation, we replace the critic's terrain encoder with a lightweight Vision Transformer (ViT) that tokenizes the heightmap, applies multi-head self-attention, and aggregates the result through a learnable `[CLS]` token—all within roughly 53K parameters.

Our contributions are:

- A ViT-enhanced critic that encodes terrain heightmaps via

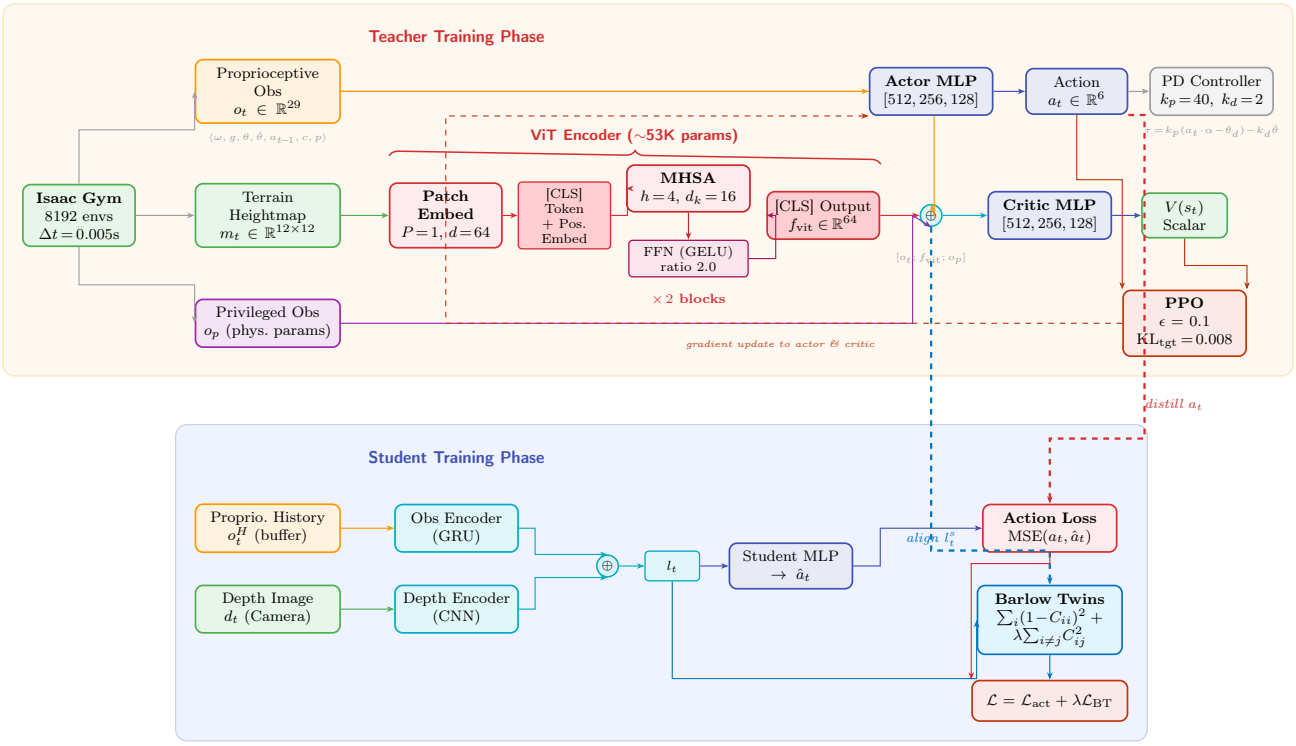


Fig. 2: Overview of the proposed ViT-enhanced teacher-student RL framework. Top: during teacher training, the 12×12 terrain heightmap is processed by the ViT-Enhanced Terrain Encoder, which extracts a spatial terrain feature $\boldsymbol{f}_{\text{vit}}$ via self-attention. This feature is concatenated ($\oplus$) with proprioceptive and privileged observations to form the critic input for value estimation $V(s_t)$. The actor MLP uses only proprioceptive observations. Bottom: during student training, only depth images and proprioceptive history are available; the student is trained via action loss and Barlow Twins feature alignment.

self-attention, providing spatially-aware value estimates at negligible parameter overhead.
- A reward function tailored to point-foot bipedal robots, combining foot trajectory tracking, gait phase synchronization, and terrain-adaptive height penalties.
- Simulation experiments on five terrain types and real-world deployment on a Limx Dynamics P1 robot, with 5.0% lower velocity error and 8.5% lower height error than the VMTS baseline.

II. Related Work

A. Reinforcement Learning for Legged Locomotion

RMA [7] introduced an adaptation module that infers latent environment parameters from proprioceptive history, enabling rapid online adaptation. Massively parallel simulation [11] further accelerated training to the order of minutes. A complementary line of work focuses on policy generalization: Walk These Ways [10] showed that a single policy with multiplicity of behavior can cover diverse locomotion tasks. On bipedal platforms specifically, rapid motor adaptation [12] and gait-conditioned policies [13] have been explored, though both rely solely on proprioception and therefore cannot anticipate terrain changes ahead of the robot.

B. Vision-Based Locomotion with Teacher-Student Learning

The teacher-student paradigm was popularized by Miki et al. [8], who used online-generated height maps as the privileged input for the teacher. Subsequent work replaced online map generation with direct depth-image input [15] and explored concurrent training of both networks [9]. VMTS [14] advanced this line by introducing an MoE model together with Barlow Twins [16] feature alignment for distillation. Outside of RL-based locomotion, Gaussian process models have been used for real-time traversability assessment [26] and safe navigation with control barrier functions [28], while efficient planners have been developed for car-like [30] and tractor-trailer robots [29] on uneven terrain, and even for maritime route planning [27]. A common limitation across RL-based methods is that terrain is either flattened into a vector (MLP) or routed through discrete experts (MoE), neither of which explicitly models the *spatial layout* within the heightmap.

C. Vision Transformers in Robotics

Since Dosovitskiy et al. [17] showed that pure Transformer architectures rival CNNs on image recognition, ViT-style models have been adopted in several robotics domains. Fu et al. [18] applied Transformers to whole-body manipulation, while ANYmal Parkour [19] used attention-based perception for agile quadruped navigation. In legged locomotion

specifically, extreme parkour [22] and humanoid parkour [23] leverage vision pipelines for obstacle negotiation, and PIE [24] combines implicit and explicit terrain representations. Wang et al. [25] further investigated which state variables matter most for robust humanoid walking. Despite this progress, all these methods apply vision modules to the *policy* (actor) side. To our knowledge, no prior work has used a ViT encoder specifically on the *critic* side to process terrain heightmaps for value estimation. Our work fills this gap: by placing a lightweight ViT inside the critic, we improve value estimates without adding complexity to the deployed student policy.

III. Method

A. System Overview

As illustrated in Fig. 2, our framework follows the teacher-student RL paradigm. During teacher training, the teacher policy receives privileged observations $\boldsymbol{s}_t^{\text{priv}} = \langle o_t, o_p, m_t \rangle$, where o_t denotes proprioceptive observations, o_p denotes physical parameters, and m_t denotes the terrain heightmap. The actor network (standard MLP) maps these observations to actions. The critic incorporates a ViT encoder to process m_t, producing a spatially-aware terrain representation for value estimation. The policy is optimized using PPO [20] with conservative clipping. During student training, the student receives only proprioceptive data and depth images, and is trained to imitate the teacher.

B. Observation Space

The proprioceptive observation $o_t \in \mathbb{R}^{29}$ comprises the angular velocity $\boldsymbol{\omega}_t$ and projected gravity vector $\boldsymbol{g}_t$ from the IMU, joint positions $\boldsymbol{\theta}_t$, joint velocities $\dot{\boldsymbol{\theta}}_t$, previous action $\boldsymbol{a}_{t-1}$, user commands $\boldsymbol{c}_t \in \mathbb{R}^3$ (desired v_x, v_y, and yaw rate), and the reference contact phase $\boldsymbol{p}_t \in \mathbb{R}^2$ (sin/cos of the gait clock):

$$o_t = \langle \boldsymbol{\omega}_t, \boldsymbol{g}_t, \boldsymbol{\theta}_t, \dot{\boldsymbol{\theta}}_t, \boldsymbol{a}_{t-1}, \boldsymbol{c}_t, \boldsymbol{p}_t \rangle. \tag{1}$$

This gives 3+3+6+6+6+3+2 = 29 dimensions. The reference contact phase is generated from a sinusoidal gait clock with period $T = 0.5$ s.

The privileged observation extends o_t with the 12×12 local terrain heightmap $m_t \in \mathbb{R}^{144}$ sampled within a $1\,\text{m} \times 1\,\text{m}$ area, the base linear velocity, foot heights, base height, friction coefficients $\mu \in [0.2, 1.6]$, motor strength scaling factors, and link masses—information that is available in simulation but not on the real robot.

C. ViT-Enhanced Critic Architecture

The critic processes the privileged observation by separating it into three components: proprioceptive observations (29 d), the terrain heightmap (144 d), and the remaining privileged features. Fig. 3 illustrates the architecture.

1) Patch Embedding: The heightmap m_t is reshaped into a 12×12 grid and partitioned into non-overlapping patches of size $P \times P$. Each patch is projected into an embedding of dimension d:

$$\boldsymbol{e}_i = \boldsymbol{W}_{\text{proj}} \boldsymbol{x}_i + \boldsymbol{b}_{\text{proj}}, \quad i = 1, \ldots, N, \tag{2}$$

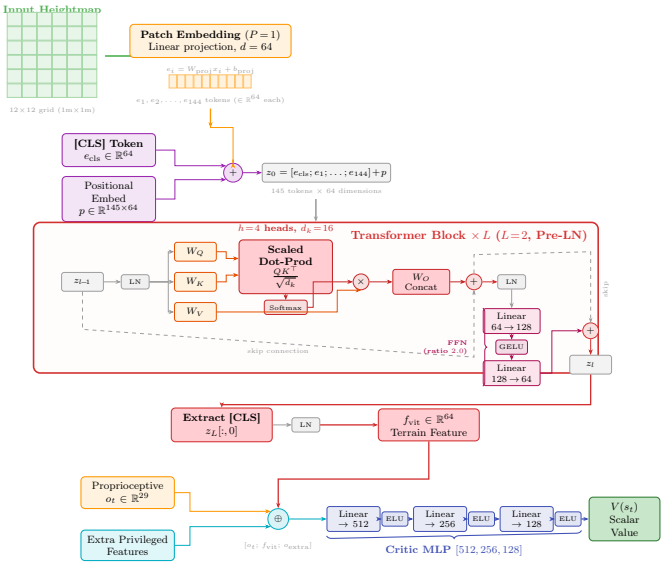


Fig. 3: ViT-enhanced critic architecture. The 12×12 heightmap is tokenized, processed by Transformer blocks, and the `[CLS]` output is concatenated with proprioceptive and extra features for value estimation.

where $\boldsymbol{x}_i \in \mathbb{R}^{P^2}$ is the flattened patch and $N = (12/P)^2$. With $P = 1$, each heightmap cell becomes an individual token ($N = 144$).

2) Positional Encoding and `[CLS]` Token: A learnable `[CLS]` token $\boldsymbol{e}_{\text{cls}} \in \mathbb{R}^d$ is prepended to the patch sequence, and positional embeddings $\boldsymbol{p}_i$ encode spatial positions:

$$\boldsymbol{z}_0 = [\boldsymbol{e}_{\text{cls}}; \boldsymbol{e}_1; \ldots; \boldsymbol{e}_N] + [\boldsymbol{p}_0; \boldsymbol{p}_1; \ldots; \boldsymbol{p}_N]. \tag{3}$$

3) Transformer Blocks: The sequence $\boldsymbol{z}_0$ is processed by L Transformer blocks with pre-normalization:

$$\boldsymbol{z}'_l = \text{MHSA}(\text{LN}(\boldsymbol{z}_{l-1})) + \boldsymbol{z}_{l-1}, \tag{4}$$

$$\boldsymbol{z}_l = \text{FFN}(\text{LN}(\boldsymbol{z}'_l)) + \boldsymbol{z}'_l, \tag{5}$$

where Multi-Head Self-Attention (MHSA) with h heads computes:

$$\text{head}_j = \text{softmax}\left(\frac{\boldsymbol{Q}_j \boldsymbol{K}_j^\top}{\sqrt{d/h}}\right) \boldsymbol{V}_j, \tag{6}$$

and the Feed-Forward Network (FFN) uses GELU activation:

$$\text{FFN}(\boldsymbol{z}) = \boldsymbol{W}_2\, \text{GELU}(\boldsymbol{W}_1 \boldsymbol{z} + \boldsymbol{b}_1) + \boldsymbol{b}_2. \tag{7}$$

We set $d = 64$, $L = 2$, $h = 4$, with FFN expansion ratio 2.0, yielding a lightweight encoder with approximately 53K parameters.

4) Feature Fusion and Value Estimation: The `[CLS]` token output $\boldsymbol{f}_{\text{vit}} = \text{LN}(\boldsymbol{z}_L)[:, 0] \in \mathbb{R}^d$ is concatenated with proprioceptive and extra features:

$$\boldsymbol{c}_{\text{in}} = [\boldsymbol{o}_{\text{proprio}}; \boldsymbol{f}_{\text{vit}}; \boldsymbol{o}_{\text{extra}}], \tag{8}$$

which is processed by a three-layer MLP $[512, 256, 128]$ with ELU activations to output the scalar value estimate $V(s_t)$.

TABLE I: Reward Function Components and Weights

Reward Term	Weight	Reward Term	Weight
Tracking lin. vel.	10.0	Feet air time	60.0
Tracking ang. vel.	4.0	Contact number	0.5
Lin. vel. (z)	-0.5	Collision	-30.0
Ang. vel. (xy)	-0.06	Feet distance	-100.0
Orientation	-6.0	Target feet ht.	-6.0
Base height	-10.0	Unbal. air time	-300.0
Joint accel.	-2.5e-7	Unbal. feet ht.	-60.0
Action rate	-0.01	Survival	0.5
Joint torque	-2.5e-5	Hip position	-2.0
Torque limits	-0.1	Contact vel.	-0.5

D. Actor Network and Action Space

The actor is a standard MLP $[512, 256, 128]$ with ELU activations, outputting a 6-dimensional action vector $\boldsymbol{a}_t$ representing target joint position deviations. Joint torques are computed through a PD controller:

$$\boldsymbol{\tau} = k_p(\boldsymbol{a}_t \cdot \alpha + \boldsymbol{\theta}_{\text{default}} - \boldsymbol{\theta}_t) - k_d\dot{\boldsymbol{\theta}}_t, \tag{9}$$

where $\boldsymbol{\theta}_{\text{default}}$ denotes the default joint positions, $k_p = 40\,\text{N·m/rad}$, $k_d = 2.0\,\text{N·m·s/rad}$, and $\alpha = 0.25$ is the action scale.

E. Training with Conservative PPO

The policy is optimized using PPO with the clipped surrogate objective:

$$\mathcal{L} = \hat{\mathbb{E}}_t\Big[\min\Big(r_t\hat{A}_t,\ \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\Big)\Big], \tag{10}$$

where r_t is the probability ratio and $\hat{A}_t$ is the advantage estimate. We adopt $\epsilon = 0.1$ (reduced from the standard 0.2) and a KL divergence target of 0.008 (reduced from 0.01). This conservative setting constrains policy update magnitudes, promoting training stability as the ViT encoder's representations evolve.

F. Student Training via Distillation

During deployment, the student policy receives only proprioceptive history $\boldsymbol{o}_t^H$ and depth images $\boldsymbol{d}_t$. An observation encoder (GRU) and a depth encoder (CNN) produce a latent representation $\boldsymbol{l}_t$, which the student MLP maps to an action $\hat{\boldsymbol{a}}_t$. The student is trained by minimizing an action imitation loss together with a Barlow Twins [16] feature alignment loss:

$$\mathcal{L}_{\text{stu}} = \text{MSE}(\boldsymbol{a}_t, \hat{\boldsymbol{a}}_t) + \lambda\mathcal{L}_{\text{BT}}(\boldsymbol{l}_t^s, \boldsymbol{l}_t), \tag{11}$$

where $\lambda = 0.5$ balances the two objectives, $\boldsymbol{l}_t^s$ is the teacher's fused latent representation, and $\mathcal{L}_{\text{BT}}$ encourages the cross-correlation matrix $\boldsymbol{C}$ between student and teacher encodings to approach the identity.

G. Algorithm Summary

Algorithm 1 summarizes the complete training procedure.

Algorithm 1 ViT-Enhanced Teacher-Student RL

Input: Terrain heightmap $\boldsymbol{m}_t$, proprioceptive obs $\boldsymbol{o}_t$, privileged obs $\boldsymbol{o}_p$
Output: Trained student policy π_{stu}

// Phase 1: Teacher Training
Initialize actor π_ϕ, ViT-critic V_ψ, ViT encoder f_{vit}
for iteration $= 1, \ldots, N_{\text{tea}}$ **do**
  Collect trajectories with π_ϕ in 8192 parallel envs
  $\boldsymbol{f}_{\text{vit}} \leftarrow \text{ViTEncoder}(\boldsymbol{m}_t)$ *// [CLS] + Self-Attn*
  $\boldsymbol{c}_{\text{in}} \leftarrow [\boldsymbol{o}_t;\ \boldsymbol{f}_{\text{vit}};\ \boldsymbol{o}_p]$ *// Feature fusion*
  $V(s_t) \leftarrow \text{CriticMLP}(\boldsymbol{c}_{\text{in}})$
  $\boldsymbol{a}_t \leftarrow \pi_\phi(\boldsymbol{o}_t)$ *// Actor uses proprio only*
  Compute advantages $\hat{A}_t$ with GAE using $V(s_t)$
  Update ϕ, ψ via PPO ($\epsilon = 0.1$, $\text{KL}_{\text{tgt}} = 0.008$)
  Update terrain curriculum based on performance
end for

// Phase 2: Student Training
Initialize student encoder g_ξ, student MLP π_{stu}
for iteration $= 1, \ldots, N_{\text{stu}}$ **do**
  Collect data using teacher π_ϕ (frozen)
  $\boldsymbol{l}_t^s \leftarrow [\boldsymbol{o}_t;\ \boldsymbol{f}_{\text{vit}};\ \boldsymbol{o}_p]$ *// Teacher latent*
  $\boldsymbol{l}_t \leftarrow g_\xi(\boldsymbol{o}_t^H, \boldsymbol{d}_t)$ *// Student latent*
  $\hat{\boldsymbol{a}}_t \leftarrow \pi_{\text{stu}}(\boldsymbol{l}_t)$
  $\mathcal{L}_{\text{act}} \leftarrow \text{MSE}(\boldsymbol{a}_t, \hat{\boldsymbol{a}}_t)$
  $\mathcal{L}_{\text{BT}} \leftarrow \text{BarlowTwins}(\boldsymbol{l}_t^s, \boldsymbol{l}_t)$
  Update ξ via $\mathcal{L} = \mathcal{L}_{\text{act}} + \lambda\mathcal{L}_{\text{BT}}$
end for
return π_{stu}

H. Reward Function Design

We design a comprehensive reward function $r_t = \sum_i w_i r_i(s_t, a_t)$. Key components are detailed in Table I.

Velocity Tracking. The linear velocity tracking reward uses an exponential kernel:

$$r_{\text{lin}} = \exp\big(-\|\boldsymbol{v}_{xy}^{\text{cmd}} - \boldsymbol{v}_{xy}\|^2/\sigma\big), \tag{12}$$

with $\sigma = 0.25\,\text{m}^2/\text{s}^2$, chosen so that the reward drops to ~ 0.37 at a 0.5 m/s tracking error.

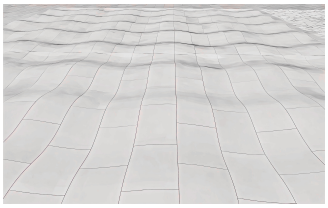
Gait Phase Synchronization. A sinusoidal reference contact with period $T = 0.5\,\text{s}$ determines the expected foot contact states. The reward evaluates whether actual contacts match the gait phase:

$$r_{\text{phase}} = \frac{1}{2}\sum_{i=1}^{2}\left[\mathbb{1}[c_i = \hat{c}_i] - \mathbb{1}[c_i \neq \hat{c}_i]\right]. \tag{13}$$

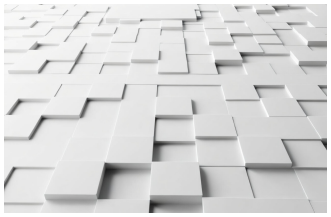
Foot Trajectory Control. To ensure smooth foot trajectories, target foot heights follow a cosine profile:

$$h_i^{\text{tgt}} = \tfrac{1}{2}\max(h^{\max}(1 - \cos(4\pi t/T)),\ 0) \cdot \bar{C}_i, \tag{14}$$

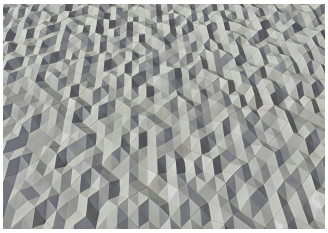
where $h^{\max}$ is the maximum foot height and $\bar{C}_i$ is the complement of the reference contact phase.



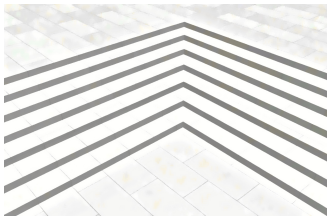
(a) Smooth slopes



(b) Rough slopes



(c) Discrete obstacles



(d) Ascending stairs

Fig. 4: Representative simulated terrain types used for curriculum-based training across five difficulty categories.

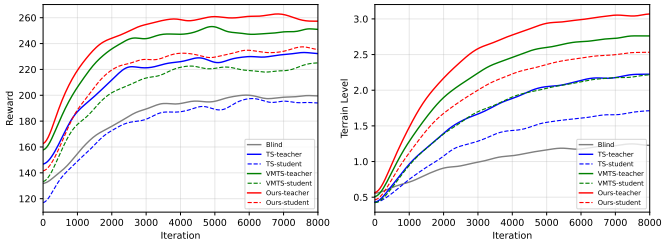


Fig. 5: Training curves. Left: reward over iterations. Right: terrain level progression. Our method reaches level 3.1, compared to 2.8 (VMTS) and 2.3 (TS).

TABLE II: Quantitative Simulation Results (mean±std over 3 seeds). Velocity error and height error are in m/s and m, respectively.

Method	**Vel. Error ↓**	**Ht. Error ↓**	**Survival (s) ↑**
Blind	0.929±0.041	0.076±0.008	84.68±1.23
TS	0.673±0.032	0.097±0.011	88.60±0.85
VMTS [14]	0.535±0.025	0.071±0.006	88.49±0.78
Ours	**0.508±0.022**	**0.065±0.005**	**89.21±0.64**

IV. Experimental Evaluation

A. Simulation Setup

We employ NVIDIA Isaac Gym [21] with 8192 parallel robots. The simulation runs at $\Delta t = 0.005\,\text{s}$ with decimation 4, yielding 50 Hz policy frequency. Episodes last 20 s, and training runs for 8000 iterations with 24 steps per iteration.

Terrain Configuration. A curriculum-based environment contains five terrain types in equal proportions (20% each): smooth slopes, rough slopes, ascending stairs, descending stairs, and discrete obstacles. The terrain grid has 10 difficulty levels $\times$ 20 terrain columns, each patch spanning $8 \times 8\,\text{m}$.

Domain Randomization. To facilitate sim-to-real transfer, we randomize friction coefficients $[0.2, 1.6]$, payload mass $[-1, 2]$ kg, center-of-mass offset $[\pm 0.03, \pm 0.02, \pm 0.03]$ m, and external pushes (every 7 s, max 0.5 m/s).

Baselines. We compare against three approaches under identical reward and training conditions:

- **Blind**: Proprioceptive-only MLP actor-critic.
- **Teacher-Student (TS)**: Two-phase framework with MLP critic receiving the heightmap as a flat vector.
- **VMTS** [14]: MoE-based teacher-student with Barlow Twins alignment.

Fig. 4 shows the four representative terrain types used in the curriculum.

B. Training Curves

Fig. 5 shows reward and terrain level over training iterations. All vision-based methods converge to similar reward plateaus, but the terrain level—an indicator of the hardest terrain a policy can reliably traverse—differs substantially. Our method reaches level 3.1, compared to 2.8 for VMTS and 2.3 for TS. The gap is largest during iterations 3000–5000, precisely when the curriculum shifts to harder terrains (stairs, discrete obstacles), suggesting that self-attention over the heightmap provides a richer gradient signal for the critic when terrain geometry becomes complex.

C. Quantitative Evaluation

We deploy 128 robots across multiple terrains with fixed proportions and measure velocity tracking error, height tracking error, and average survival time within 100 s. Results are presented in Table II.

Without terrain input (Blind), velocity error is 0.929 m/s and survival drops to 84.68 s—the robot frequently falls on stairs and rough slopes. Adding the heightmap as a flat vector (TS) already helps (velocity error 0.673), but the MoE routing in VMTS brings a further 20% reduction (0.535). Our ViT critic pushes velocity error to 0.508 (5.0% below VMTS) and height error to 0.065 m (8.5% below VMTS). The improvement is consistent across seeds (± 0.022 vs. ± 0.025), indicating that the gain comes from the architecture rather than lucky initialization. We attribute this to the self-attention mechanism's ability to jointly reason about neighboring heightmap cells, something neither flat MLPs nor discrete MoE routing can achieve.

D. Ablation Study

Table III isolates the effect of each design choice. The largest degradation occurs when the ViT encoder is replaced with a standard MLP: velocity error rises by 12.8% and height error by 26.2%, directly supporting our hypothesis that spatial attention over the heightmap matters. Switching from conservative clipping ($\epsilon = 0.1$) to the standard $\epsilon = 0.2$ yields a smaller but consistent drop, which we attribute to larger policy updates destabilizing the co-evolving ViT representations. Coarsening the tokenization ($P = 2$, 36 tokens) or reducing depth ($L = 1$) each degrade performance moderately, confirming that both fine spatial resolution and sufficient attention depth contribute.

TABLE III: Ablation Study Results (mean±std over 3 seeds)

Variant	Vel. Err. ↓	Ht. Err. ↓	Surv. (s) ↑
Ours (full)	**0.508±0.022**	**0.065±0.005**	**89.21±0.64**
w/o ViT (MLP)	0.573±0.028	0.082±0.007	88.30±0.91
w/o cons. clip	0.537±0.024	0.072±0.006	88.76±0.72
Patch size $P=2$	0.524±0.023	0.070±0.006	88.92±0.68
Depth $L=1$	0.548±0.026	0.078±0.007	88.61±0.83

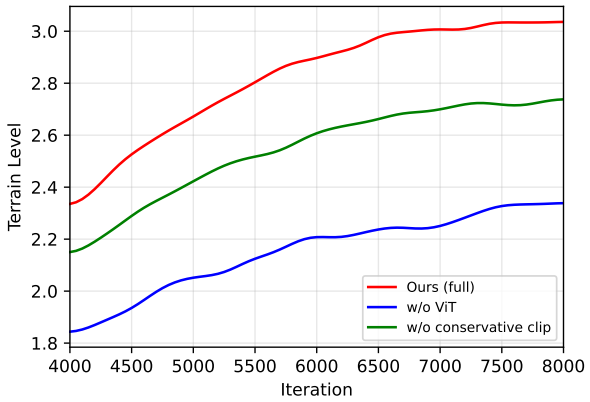


Fig. 6: Terrain level curves for ablation variants during student training (iterations 4000–8000). Removing either the ViT encoder or conservative clipping causes ∼0.3 level drop.

Fig. 6 plots terrain level during the student phase. The full method maintains a clear lead; removing either the ViT encoder or conservative clipping causes the curve to plateau roughly 0.3 levels lower, and the two ablations compound—confirming that both components are necessary.

E. Simulation Locomotion Results

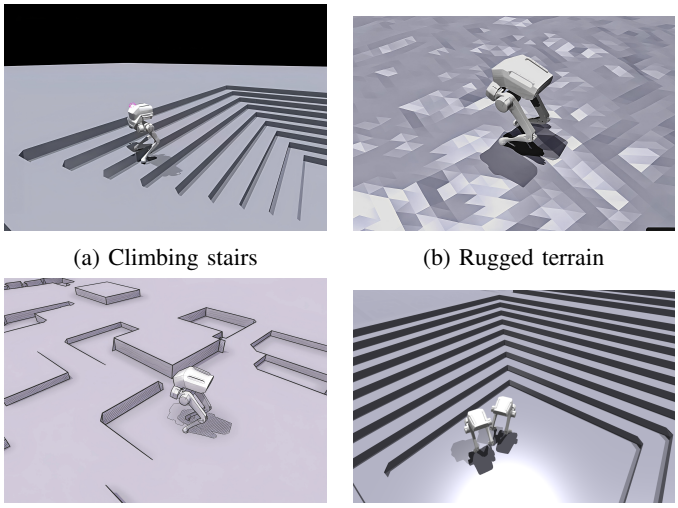
Fig. 7 shows simulation snapshots. On stairs, the robot lifts its feet noticeably higher than on flat ground—a behavior that was not hand-coded but emerged from the ViT critic's ability to assign lower value to states where foot clearance is insufficient relative to the upcoming stair edge. On rough and uneven terrain, the gait becomes wider and more cautious, again without explicit gait switching logic.

F. Real-World Experiments

We deploy the trained student policy on a Limx Dynamics P1 bipedal robot, which has 6 actuated joints (3 per leg: hip, knee, and ankle) and is equipped with an IMU and an Intel RealSense D435i depth camera. The policy runs at 50 Hz on the onboard computing unit.

Stair Climbing. As shown in Fig. 8, the robot ascends and descends a staircase with approximately 15 cm step height. The increased foot clearance learned in simulation transfers to the real robot without fine-tuning: the student policy raises its swing foot ∼3 cm higher on stairs than on flat ground. The stair-up and stair-down sequences each last approximately 8 s over 5 steps, with zero falls across 10 consecutive trials.

Multi-Terrain Traversal. On flat indoor tile, outdoor rough concrete, and a 10° inclined ramp, the robot walks at a commanded velocity of 0.3 m/s with an average tracking error below 0.08 m/s (measured over 30 s per surface). No fall occurred during any trial, though we note that the tested terrain variety is more limited than the five simulation categories.



(a) Climbing stairs (b) Rugged terrain

(c) Uneven terrain (d) Stair bottom approach

Fig. 7: Simulation snapshots of the bipedal robot navigating various terrains with the trained policy.



(a) Ascending stairs (b) Descending stairs

Fig. 8: Real-world deployment on a bipedal robot. The robot successfully climbs and descends stairs with approximately 15 cm step height.

V. Conclusion

We showed that replacing the critic's terrain encoder with a lightweight ViT (53K parameters, $d = 64$, $L = 2$) yields measurable gains in a teacher-student bipedal locomotion pipeline: 5.0% lower velocity error and 8.5% lower height error than the MoE-based VMTS baseline across five terrain types, with successful real-world stair climbing on a Limx P1 robot.

Limitations. The improvement over VMTS, while consistent, is incremental. The real-world experiments are limited to three surface types and lack long-duration outdoor tests. Additionally, the ViT encoder processes a 12×12 heightmap—coarser resolutions or larger maps may require architectural changes.

Future Work. Two directions seem most promising: (1) moving the ViT encoder to the *actor* side so that the deployed student can directly benefit from spatial attention over depth features, and (2) combining the ViT critic with a trajectory planner for more aggressive locomotion tasks such as gap jumping.

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